GUIDELINES FOR

Evaluating Water in Pit Slope Stability

EDITORS: GEOFF BEALE AND JOHN READ

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INTRODUCTION

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1 Scope of LOP project hydrogeological studies

The presence of water has a detrimental effect on slope stability. Water pressure acting in the pore spaces, fractures or other discontinuities in the materials that make up the pit slope will reduce the strength of those materials, and may therefore have a large influence on the performance, safety and economics of a mining operation.

The past 10 to 15 years has seen a significant improvement in the understanding of mining hydrogeology. Most mines now incorporate groundwater and surface water programs at all stages of the mining cycle. The programs are driven by both engineering and mine economics, and by increasing regulatory constraints, environmental laws and safety awareness. However, there remains a tendency to apply over-simplified approaches for assessing the distribution of pore water pressures in closely jointed rock masses and other pit slope materials. There is increasing awareness that an inadequate understanding of pore water pressures leads to mine production losses, either from conservatively over-designed slopes (i.e. slopes that are flatter than necessary), or under-designed slopes that fail.

In response to these issues, in 2009, the LOP project commenced a number of detailed hydrogeological studies to assess the effect of pore water pressures on the stability of jointed rock. The studies were designed to develop an understanding of the flow process in rock masses at different scales, particularly those with poor permeability and/or limited hydraulic connectivity between fractures, joints and other discontinuities. The objective was to develop a methodology that would help geotechnical design engineers decide how to treat water pressures in their pit walls.

The primary considerations for the studies have been as follows.

- The scale at which the water pressures are important for reducing rock strength, and how this relates to the grid and mesh sizes used in most geotechnical analyses.
- Whether water pressures are important throughout the rock mass or only in the major discontinuities.
- Whether analysis of the water pressure distribution can consider the rock mass as an equivalent continuum, so the flow analyses can employ some form of equivalent porous-medium approach.
- How slope deformation and dilation of the materials may change the water pressures within the rock mass and what are the resulting changes in the permeability of the fracture network.

For the geotechnical design engineer, a fundamental decision is whether to compute the water pressures only in the individual structures that cut through the rock mass; whether to compute the water pressures only in the intact rock within the intervening rock bridges (including the mélange of lesser joints, microfabric and intact rock substance included within the rock bridges); or whether to compute them in both.

To address these considerations, the hydrogeological studies focused on three core questions.

1. ***How does a fractured rock slope depressurise***, which sought to define the role and the response of the various orders of fractures within the rock mass by:
   a. reviewing the literature from other industries with interests in the same or similar questions (e.g. nuclear, hydrocarbon and construction);
   b. analysing the structural and hydrogeological data from a number of large open pit mines to develop a conceptual model of how depressurisation occurred in fractured rock pit slopes for different rock types, alteration zones and structural settings, and in a range of different mine site environments.
2. ***How should depressurisation be simulated numerically***, which sought to formulate a standardised

modelling approach for water pressure simulation for a range of environments and settings, with guidelines for determining:

a. the approach that should be used to construct, calibrate and run different types of numerical models;
b. the selection of appropriate model codes and procedures for each type of hydrogeological setting;
c. when it is appropriate to use an equivalent porous-medium flow code;
d. when it is necessary to use a fracture flow code;
e. whether to use a transient or steady-state modelling approach;
f. when it is necessary to use a coupled hydromechanical modelling approach;
g. how active drainage measures (e.g. wells, drains, tunnels) should be simulated within the model.

3. ***How should pore water pressures be incorporated into the slope design model***, which sought to outline recommended procedures and formats for providing basic water pressure distributions and responses into limit equilibrium and numerical slope stability programs, and assessing the impact of water pressures on rock mass strength.

Initially, it was intended that the information obtained from these three questions would form the basis of a handbook detailing the hydrogeological procedures that should be followed when performing open pit slope stability design studies. However, it was realised that such a handbook would not be complete without supporting sections that outlined how water pressures are characterised and reported, how slope depressurisation systems are implemented and maintained, and how predicted versus actual conditions and behaviour are monitored and reconciled.

Following the initial research using the data from Diavik and the other case study sites, it also quickly became apparent that the wider site-scale hydrogeology (and, in particular, groundwater recharge) is the most important factor for the assessment of water in the pit slope design. The scope of these guidelines was therefore further widened to include the characterisation and assessment of the wider site-scale hydrogeology. Furthermore, for a large number of pits, it is not possible to achieve the required pit slope pore pressure goals without the operation of an efficient general mine dewatering system. The general principles for modelling and design of mine dewatering systems are therefore also included.

The book was specifically designed to follow up on Chapter 6, (the Hydrogeological Model) in the LOP project's book *Guidelines for Open Pit Slope Design* (Read & Stacey, 2009). The contents are divided into the six sections shown in Figure 1 (The hydrogeological assessment process), starting with a section that outlines

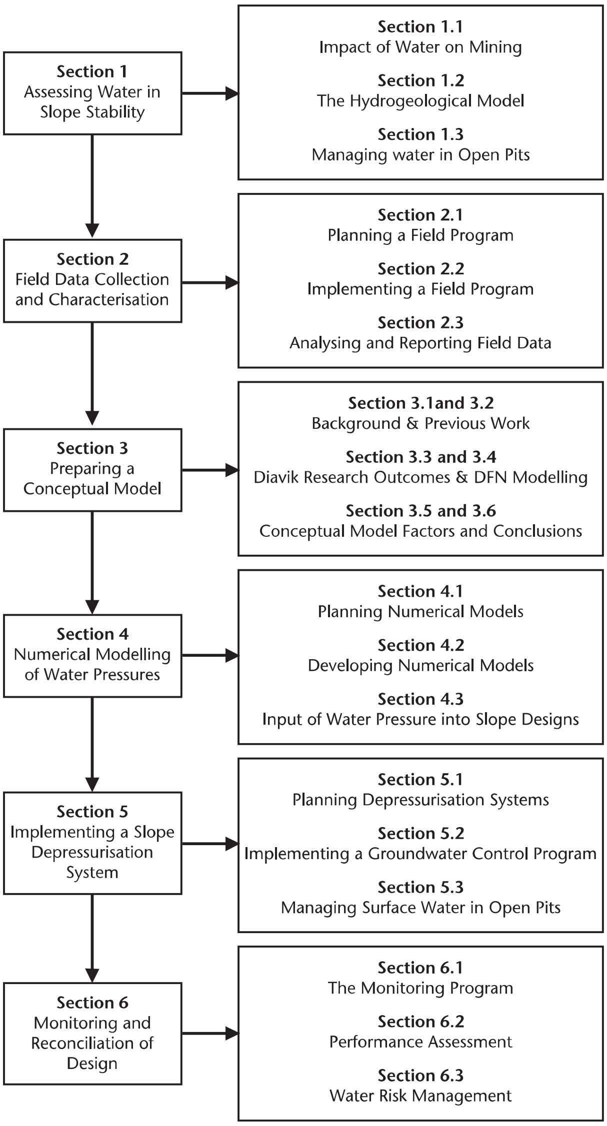

Figure 1: The hydrogeological assessment process

the framework used to assess the effect of water in slope stability (Section 1), followed by sections outlining how the water pressures are measured and tested in the field (Section 2), how a conceptual hydrogeological model is prepared (Section 3), how water pressures are modelled numerically (Section 4), how slope depressurisation systems are implemented (Section 5), and how the performance of a slope depressurisation program is monitored and reconciled with the design (Section 6).

The hydrogeological pathway followed through the slope design process (Figure 2) is illustrated in Figure 3. Sections 1, 2 and 3 are part of the Hydrogeology and Geotechnical Model sections of the process, Section 4 reports to the Groundwater Pressures and Stability Analyses sections, and Sections 5 and 6 relate to the Implementation section (general mine dewatering, slope depressurisation and monitoring).

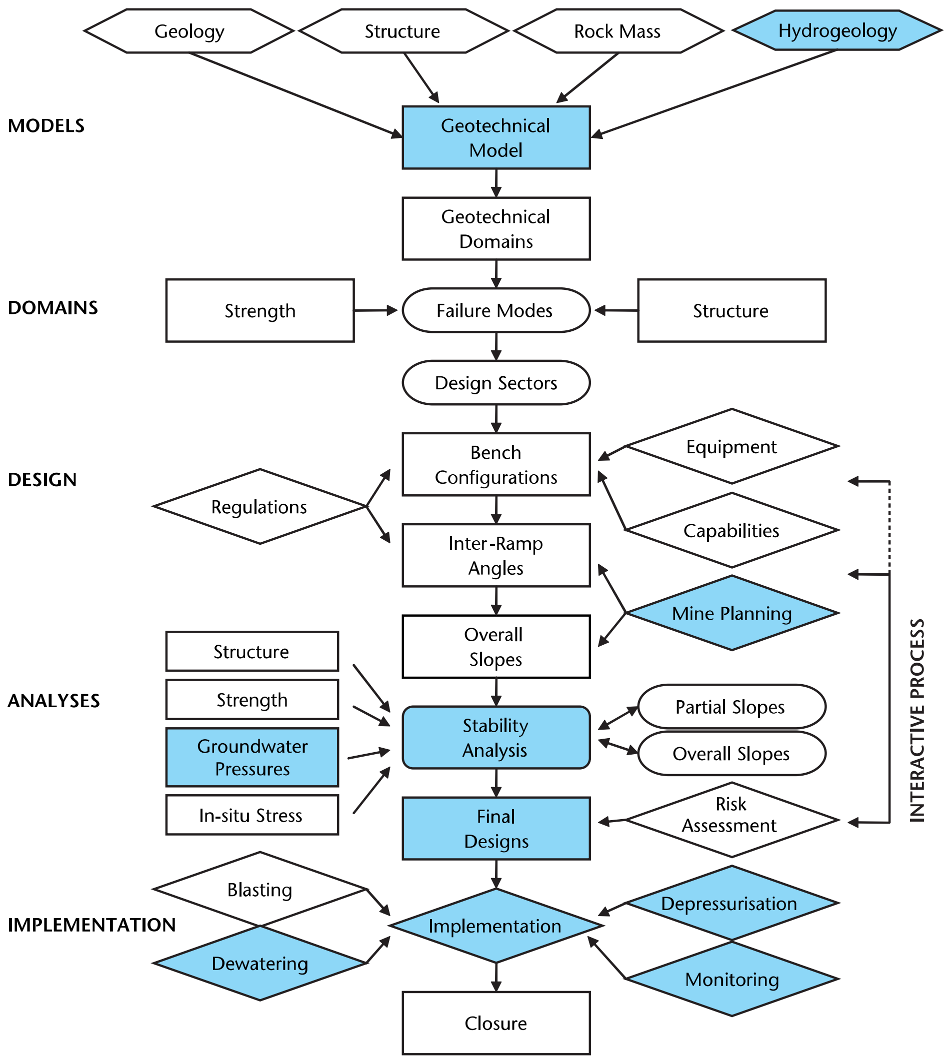

Figure 2: The slope design process, highlighting hydrogeologically important steps

When implementing the actions outlined in Figures 1 and 2, it is important to recognise that they are not completed just once but are required at every stage of the evaluation of a mineral deposit, from the initial conceptual and scoping-level designs through to the pre-feasibility, feasibility, final designs and expansion phases for an operating pit. They may also need to be considered for maintaining stability during mine closure and for long-term post-closure of the mine.

The level of effort required at each stage of project development is shown for each component of the geotechnical model in Table 1. As the developing project progresses from level to level, so there should be a corresponding increase in the level of confidence in the data. Target levels of data confidence are shown at the bottom of Table 1.

The target levels of data confidence shown in Table 1 are subjective, but were developed by the sponsors of the LOP project to provide guidelines to the level of certainty required at each stage of project development. They can also be used to indicate the level of expenditure that may be required. For example, for the geological model, the greatest level of effort and expenditure will occur early in project development, with quite high levels of confidence being obtained by Level 2. In contrast, for the hydrogeological model, the greatest level of effort and expenditure will be at Levels 3 and 4, with confirmatory work being required to ensure there is the required level of confidence for the detailed slope design studies and the planning of any required depressurisation systems.

When implementing the agreed final slope design, it is vital that the geotechnical, mine planning and mine operations engineers understand the need to integrate the water monitoring system with the mine operations activities on a day-to-day basis from Day 1 of the project. Poorly organised communications between the designers and planners and mine operations will interfere with the organised collection of data, diminishing the value and

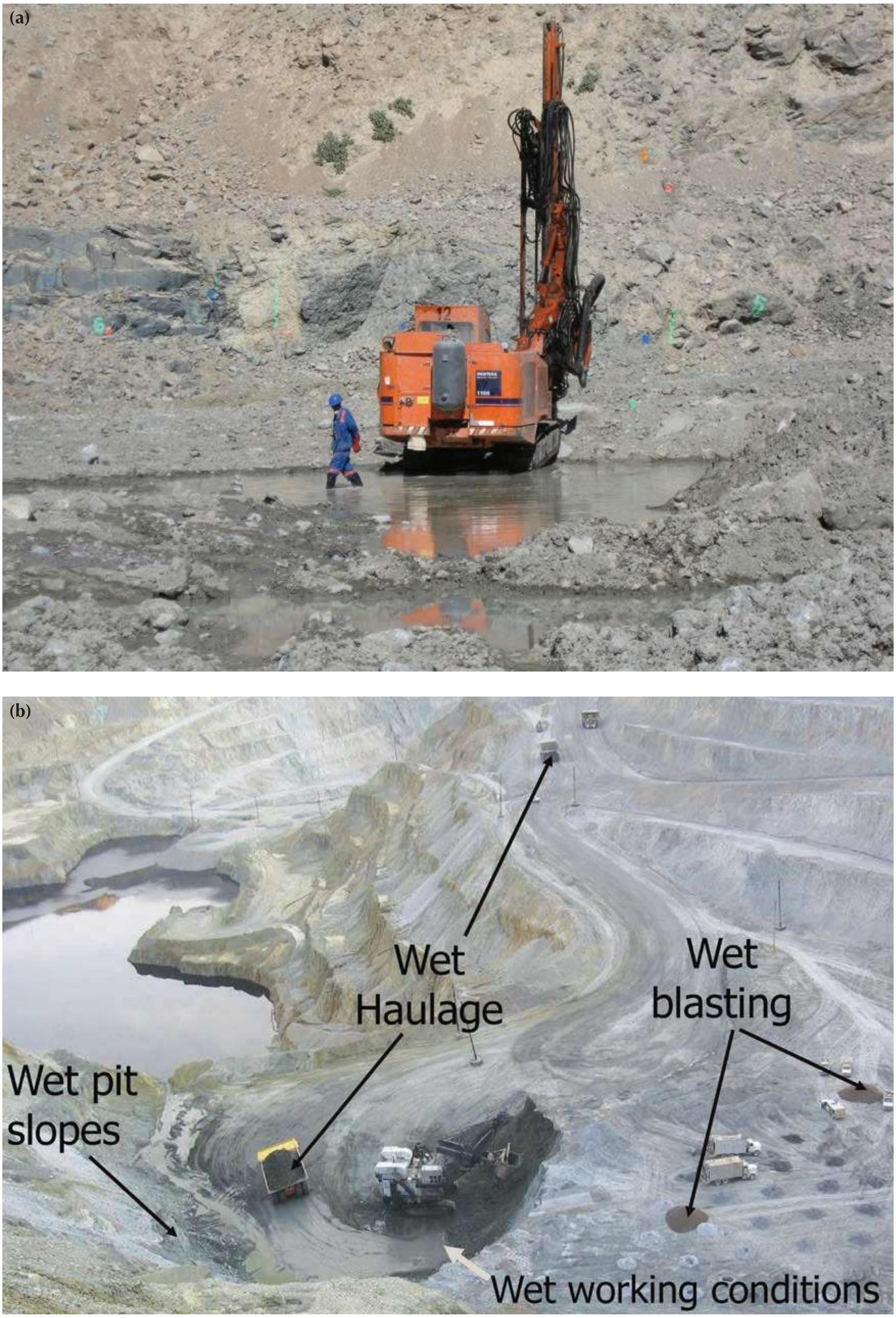

Figure 4: Adverse effect of pit floor inundation on operations and safety (Source: Schlumberger Water Services)